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A Seminar Report on

Sign Language to Text Translator

Submitted in partial fulfilment for the award of
THIRD YEAR DIPLOMA IN
COMPUTER ENGINEERING
By

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FOR ACADEMIC YEAR : 2026– 2027

Department of Computer Engineering

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(Institute Code : 0059)

S.S.V.P.S's Bapusaheb Shivahirao Deore Polytechnic , Dhule

(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMENT OF COMPUTER ENGINEERING

Certificate

This is to certify that,

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Name : Prof. N. D. Patel

Head of the Department

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ABSTRACT

Sign language is the main way of communication for deaf and mute people, but most people do not understand it. This creates a communication gap between the deaf/mute community and the general public, and human interpreters are costly and are not always available when they are needed.

This seminar report studies a Sign Language to Text Translator that recognises hand gestures in real time using a normal webcam. Video is captured with OpenCV, key hand landmarks are extracted using Google MediaPipe (21 three-dimensional landmarks per hand, 126 values per frame for two hands), and a Long Short-Term Memory (LSTM) network studies a sliding sequence of 30 frames to recognise moving gestures. The recognised gesture is shown as readable text and can also be spoken using the offline Text-to-Speech library pyttsx3. The report reviews six related research works, explains the working principle and system architecture, lists the advantages and disadvantages, and presents the project initiation details such as problem statement, scope, objectives, stakeholders, execution plan and resources.

Keywords: Sign Language, Gesture Recognition, MediaPipe, LSTM, OpenCV, Text-to-Speech.

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CHAPTER 1

INTRODUCTION

Communication is a basic human need, but for people who are deaf or mute, spoken language is not a practical way to communicate. Over 70 million people around the world rely on sign language, which uses hand shapes, hand movement and facial expressions to express ideas. Most people in the general public, however, never learn sign language, so a big communication gap exists between the deaf/mute community and everyone else.

Human interpreters can bridge this gap, but they are costly and are not always available in places such as schools, hospitals, banks, bus stations and service centres. Other technical solutions need expensive sensor gloves or special depth-camera hardware, which limits their use in daily life.

Computer Vision and Deep Learning can now recognise hand gestures and convert them into text. A camera captures the signs, software finds the hands and their key points in every frame, and a trained neural network decides which sign is being shown. The result can be displayed as readable text and can also be spoken aloud, so that a person who does not know sign language can understand the message immediately.

The solution studied in this report is a low-cost, AI-powered virtual interpreter that needs only a standard webcam. It uses OpenCV for video capture, Google MediaPipe for hand-landmark extraction, an LSTM network (built with TensorFlow) to recognise moving gestures from a sequence of 30 frames, and pyttsx3 for offline text-to-speech output.

The aim of this report is to explain the basic concepts, working principle, system architecture, tools, advantages and limitations of a Sign Language to Text Translator, and to identify the resources needed to build a prototype. This is an educational, literature-based seminar report and not a fully validated commercial product.

CHAPTER 2

LITERATURE REVIEW

2.1 Real-time American Sign Language Recognition Using Hidden Markov Models

Starner and Pentland (1995) presented one of the early systems for recognising American Sign Language (ASL) from video in real time. Features from the hands in the video are modelled with Hidden Markov Models (HMMs). The work showed that continuous signing can be recognised by a computer, but it used a small vocabulary and a controlled setup.

2.2 Sign Language Recognition Using Convolutional Neural Networks

Pigou et al. (2014) applied a Convolutional Neural Network (CNN) to recognise sign gestures using data from a Kinect sensor (Italian gestures). The CNN learns visual features directly from the data and gave good recognition results. The limitation is that a special depth-camera is needed, which is not available to every user.

2.3 MediaPipe Hands: On-device Real-time Hand Tracking

Zhang et al. (2020) describe MediaPipe Hands, which uses a palm detector and a hand-landmark model to give 21 three-dimensional hand landmarks in real time on an ordinary device. It is the basis of the feature extraction in this seminar. It provides only landmark positions, not the meaning of a sign, and errors can occur when hands are occluded.

2.4 Long Short-Term Memory

Hochreiter and Schmidhuber (1997) introduced the Long Short-Term Memory (LSTM) network, a recurrent neural network that can learn long-term dependencies in sequences. This makes it suitable for gestures, because a moving sign is a sequence of hand positions over time. LSTM needs enough sequence data and training time.

2.5 Neural Sign Language Translation

Camgoz et al. (2018) treated sign language translation as a neural machine translation problem. An encoder-decoder network with attention translates sign video into spoken-language text. It was trained on the RWTH-PHOENIX-Weather 2014T dataset. The approach needs large annotated data and is limited to the weather domain.

2.6 Sign Language Transformers

Camgoz et al. (2020) proposed Sign Language Transformers, which perform sign recognition and translation jointly in an end-to-end manner using the Transformer architecture, again on RWTH-PHOENIX-Weather 2014T. The method gives strong results but needs high computing power and is limited to a specific domain.

2.7 Research Gap and Project Direction

Early methods work only in controlled conditions, CNN-based methods may need special hardware, and large translation models need big annotated datasets and high computing power. There is a need for a simple system that works with an ordinary webcam on a normal laptop. Combining MediaPipe landmarks with an LSTM gives a light and fast solution,

because landmark coordinates are used instead of raw images. The proposed seminar studies such a low-cost translator.

2.8 Conclusion of Literature Review

The reviewed literature shows the progress from HMM-based recognition to CNNs, LSTMs and Transformers, together with real-time hand tracking. These findings form the foundation of this report, which studies a webcam-based Sign Language to Text Translator that uses MediaPipe landmarks and an LSTM model.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	HMM-based ASL recognition (Starner and Pentland, 1995)	Hidden Markov Models on video features	Small American Sign Language vocabulary	Real-time sign recognition from video	Small vocabulary; controlled setup
2	CNN-based gesture recognition (Pigou et al., 2014)	Convolutional Neural Network with Kinect	Kinect gesture data (Italian gestures)	Recognise sign gestures with CNN	Needs special depth-camera hardware
3	Hand tracking (Zhang et al., 2020 – MediaPipe Hands)	Palm detector and hand landmark model	Annotated hand images	Real-time 21-point 3D hand landmarks on device	Gives landmarks only, not meaning; occlusion errors
4	Sequence learning (Hochreiter and Schmidhuber, 1997)	Long Short-Term Memory (LSTM)	Sequence learning tasks	Learn long-term dependencies in sequences	Needs enough sequence data and training time
5	Neural Sign Language Translation (Camgoz et al., 2018)	Encoder-decoder network with attention	RWTH-PHOENIX-Weather 2014T	Translate sign video into spoken-language text	Needs large annotated data; weather domain only
6	Sign Language Transformers (Camgoz et al., 2020)	Transformer, joint recognition and translation	RWTH-PHOENIX-Weather 2014T	End-to-end sign recognition and translation	High computing need; limited domain

Table No. 2.1: Literature Review:

CHAPTER 3

INFORMATION

1 . Overview of Sign Language Translation:

Sign language uses hand shapes, hand movement and facial expressions to communicate. A Sign Language to Text Translator captures signs through a camera and converts them into readable text (and speech). It acts as a low-cost virtual interpreter, so deaf/mute people can communicate with others without a human interpreter.

The system performs dynamic recognition: it understands moving words, not just static alphabet letters.

2. Need for Sign Language to Text Translation:

Over 70 million people around the world rely on sign language, while very few members of the general public understand it. Human interpreters are costly and not always available, and sensor gloves or special hardware are expensive. An AI-based translator that needs only a standard webcam can reduce this communication gap in schools, hospitals and service centres.

3. Technologies and Tools Used:

The following technologies and tools are used in the Sign Language to Text Translator. Table No. 3.1 shows the purpose of each.

Technology	Purpose
Python	Core programming language for scripting and model integration.
OpenCV	Real-time video capture, frame reading and image pre-processing.
MediaPipe	Extracts 21 3D hand landmarks (126 features for two hands) from each frame.
LSTM (TensorFlow)	Deep learning network that analyses a sequence of 30 frames over time.
pyttsx3	Offline Text-to-Speech library that converts recognised text into audio.

Table No. 3.1: Technologies and Tools Used

4. Working Principle:

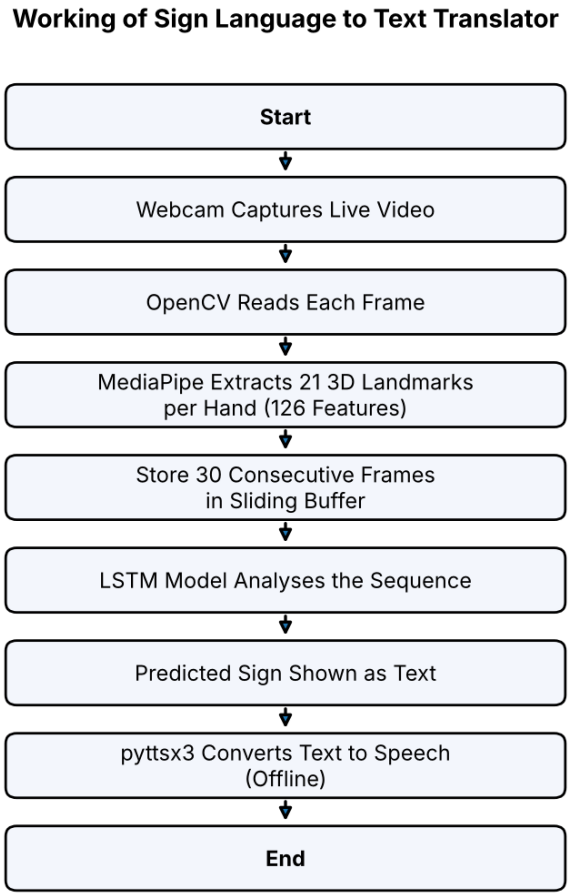


Fig 3.1: Working of Sign Language to Text Translator

The working of the Sign Language to Text Translator consists of the following steps:

1. **Capture video:** The webcam captures live video and OpenCV reads each frame.
2. **Extract landmarks:** MediaPipe detects the hands and extracts 21 3D landmarks per hand (126 features for two hands).
3. **Buffer frames:** Landmark values of 30 consecutive frames are stored in a sliding buffer.
4. **Recognise the sign:** The LSTM model analyses the sequence and predicts the sign.
5. **Display text:** The recognised sign is displayed as text on the screen.
6. **Speak the text:** pyttsx3 converts the text into speech (offline).

5. System Architecture and Main Components:

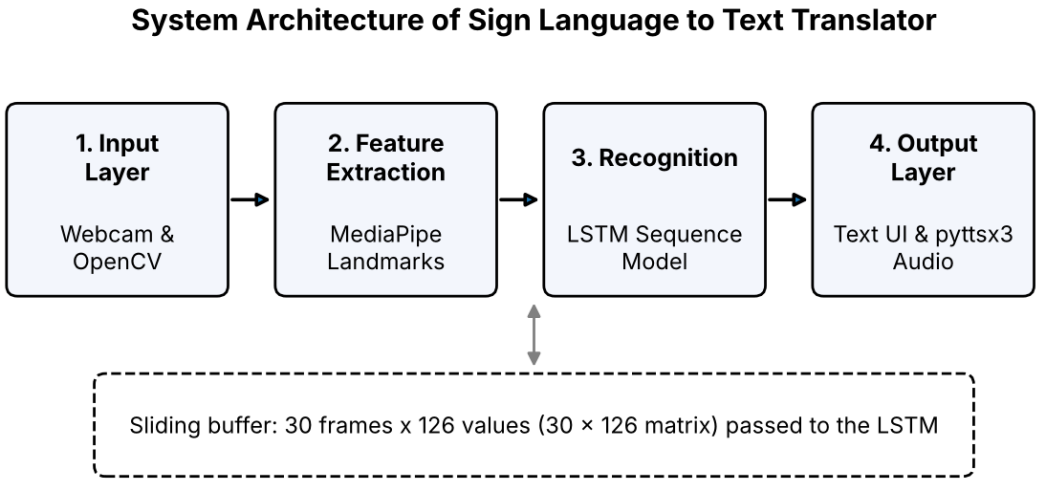


Fig 3.2: Architecture of Sign Language to Text Translator

The system has four main parts: the Input Layer (webcam and OpenCV), the Feature Extraction layer (MediaPipe landmarks), the Recognition layer (LSTM sequence model) and the Output Layer (text user interface and pyttsx3 audio).

How it processes dynamic signs: the system keeps a sliding buffer of 30 consecutive frames; each frame has 126 coordinate values. This 30 × 126 matrix is passed to the LSTM, which predicts the moving gesture.

6. Feature Extraction and Training Data:

- **Why landmarks:** Using landmark coordinates instead of raw images makes the model light, fast and less affected by the background.
- **Data collection:** Gesture sequences are recorded with the webcam for every sign to be recognised.
- **Training:** The 30-frame landmark sequences are used to train the LSTM (TensorFlow) model.
- **Testing:** Accuracy is checked on sample gestures that were not used in training.
- **Vocabulary:** The model recognises only the signs it was explicitly trained on.

CHAPTER 4

ADVANTAGES AND DISADVANTAGES

4.1 Advantages

1. **Cost-Effective:** Needs only a standard webcam, no expensive sensor gloves.
2. **Real-Time:** MediaPipe and LSTM run smoothly on normal laptop CPUs.
3. **Offline Audio:** pyttsx3 gives speech without internet.
4. **Dynamic Recognition:** Understands moving words, not just static letters.

4.2 Disadvantages

1. **Lighting Dependent:** Poor lighting reduces landmark accuracy.
2. **Hand Occlusion:** Overlapping hands can confuse tracking.
3. **Limited Vocabulary:** Recognises only the signs it was trained on.
4. **No Facial Expressions:** Ignores non-manual grammar cues.

CHAPTER 5

CONCLUSION

The Sign Language to Text Translator helps reduce the communication gap between deaf/mute people and the general public. MediaPipe landmarks combined with an LSTM model allow real-time recognition of dynamic gestures using only a webcam.

The system is low-cost and works offline, but it depends on good lighting and on a limited trained vocabulary. Hand occlusion and the absence of facial expressions are further limitations that affect accuracy.

In future, more signs, facial expressions and full sentence translation can be added, so that the system becomes a more complete virtual interpreter.

CHAPTER 6

PROJECT INITIATION

6.1 Description of problem statement :

A significant communication gap exists between the deaf/mute community and the general public. Human interpreters are costly and not always available.

Existing Systems: Many solutions need expensive sensor gloves or special hardware.

Proposed Solution: A low-cost, AI-powered virtual interpreter that recognises hand gestures through a standard webcam and converts them into text and speech.

6.2 Scope and Objectives :

The scope of the project is as follows:

1. Recognise dynamic hand gestures in real time using a webcam.
2. Convert recognised signs into text and speech.
3. Limited to the vocabulary of signs the model is trained on.
4. Out of scope: facial expressions and full sign-language grammar.

Objectives of the project:

1. Study existing sign language recognition methods.
2. Extract hand landmarks using MediaPipe and OpenCV.
3. Train an LSTM model on 30-frame gesture sequences.
4. Display the result as text and audio output.
5. Evaluate accuracy and note limitations for future work.

6.3 Stakeholders:

Stakeholders are individuals or groups who use, develop, evaluate or benefit from the system.

1. **Deaf / Mute Users:** Use the system to communicate with others.
2. **General Public:** Understand signs without learning sign language.
3. **Project Team:** Designs, develops, tests and documents the system.
4. **Schools, Hospitals and Service Centres:** Can use it where interpreters are not available.
5. **Project Guide and Evaluators:** Guide the work and evaluate design, prototype and results.

6.4 Seminar Execution Plan :

Week	Activity	Expected Output
Week 1	Selection of seminar topic and finalising the title	Topic approval
Week 2	Literature survey of sign language recognition work	Literature review notes
Week 3	Study of tools: OpenCV, MediaPipe, TensorFlow, pyttsx3	Tool list
Week 4	Designing architecture and working flow	Architecture diagram
Week 5	Collecting gesture data and building the prototype	Working prototype
Week 6	Training the LSTM and testing accuracy	Test results
Week 7	Preparing the seminar report and slides	Report and PPT draft
Week 8	Final presentation and submission	Final submission

Table No. 6.1: Seminar Execution Plan

6.5 Platform / Equipment / Resources identification :

1. Platform :

The project is developed on Windows or Linux using the Python programming language. OpenCV is used for video capture, Google MediaPipe for hand tracking, and TensorFlow / Keras to build and train the LSTM model. Visual Studio Code, Jupyter Notebook or Google Colab can be used for development and testing.

2. Equipment:

- **Computer or laptop:** For development, training and testing.
- **Webcam:** To capture live hand gestures.
- **RAM:** At least 8 GB for smooth real-time processing.
- **Internet connection:** For documentation and package downloads (the translator itself works offline).

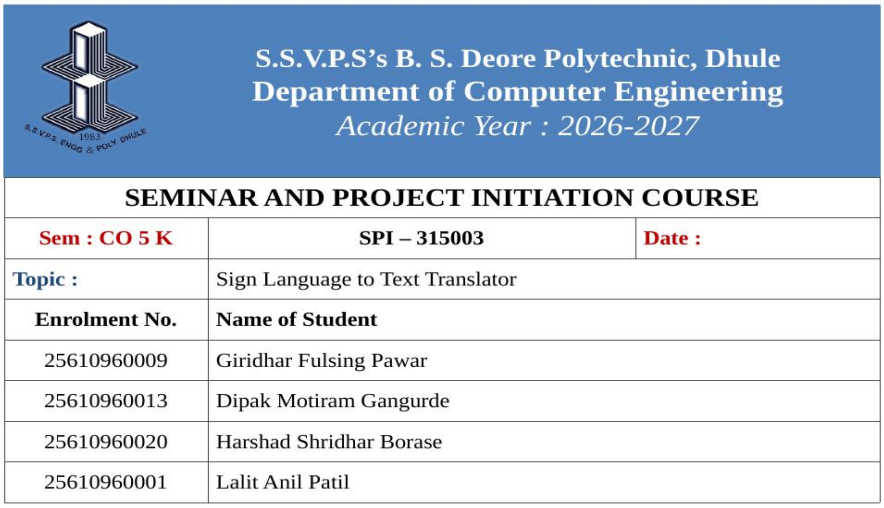
3. Resources:

Resources	Name	Purpose
Programming Language	Python	Core language for scripting and model integration.
Computer Vision	OpenCV	Video capture and pre-processing.
Hand Tracking	Google MediaPipe	Extracts 21 hand landmarks per hand.
Deep Learning	TensorFlow / Keras	Builds and trains the LSTM model.
Text-to-Speech	pyttsx3	Offline conversion of text into speech.
Data Handling	NumPy and Pandas	Handling landmark data and sequences.
Environment	VS Code, Jupyter Notebook or Google Colab	Writing, running and testing the code.
Hardware	Laptop with webcam, 8 GB RAM	Running the prototype.
Dataset	Gesture sequences recorded by the team	Training and testing for the chosen signs.

Table No. 6.2: Resources

CHAPTER 7

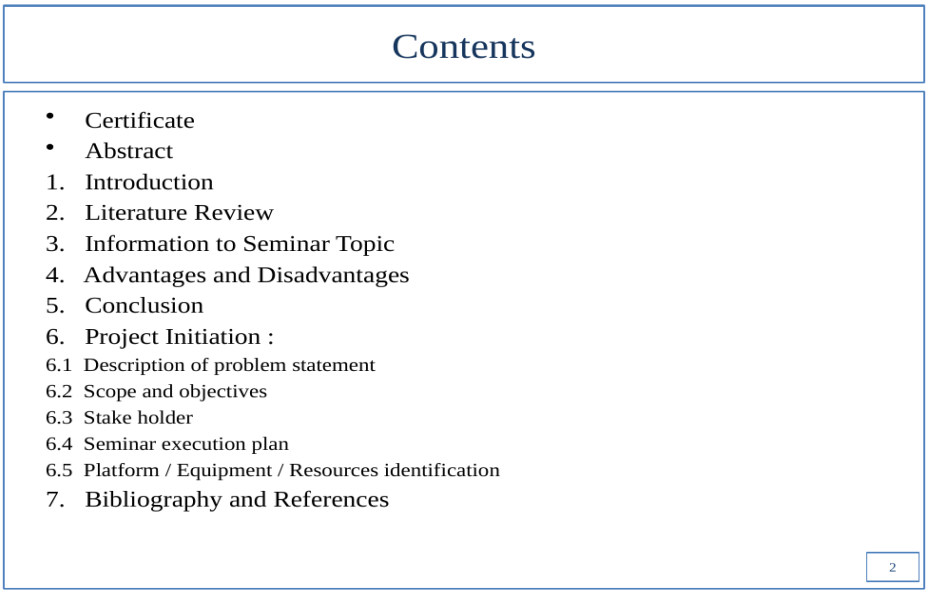
Seminar Presentation



S.S.V.P.S's B. S. Deore Polytechnic, Dhule
Department of Computer Engineering
Academic Year : 2026-2027

SEMINAR AND PROJECT INITIATION COURSE		
Sem : CO 5 K	**SPI – 315003**	**Date :**
Topic :	Sign Language to Text Translator	
Enrolment No.	**Name of Student**	
25610960009	Giridhar Fulsing Pawar	
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25610960020	Harshad Shridhar Borase	
25610960001	Lalit Anil Patil	

Fig. No. 7.1: Seminar Title Page



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3. Information to Seminar Topic
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation :

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6.2 Scope and objectives
6.3 Stake holder
6.4 Seminar execution plan
6.5 Platform / Equipment / Resources identification

7. Bibliography and References

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Fig. No. 7.2: Content Page

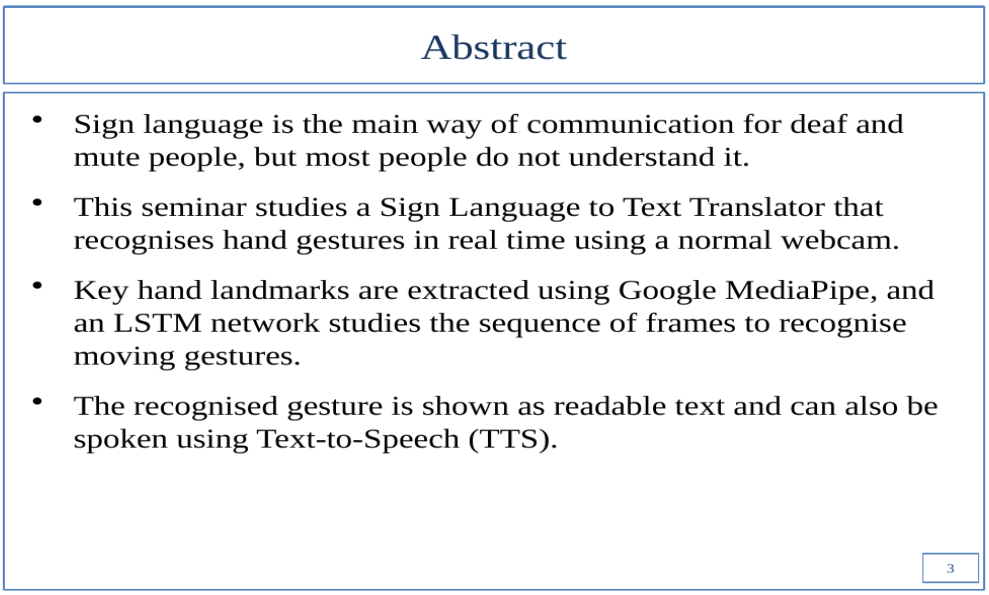


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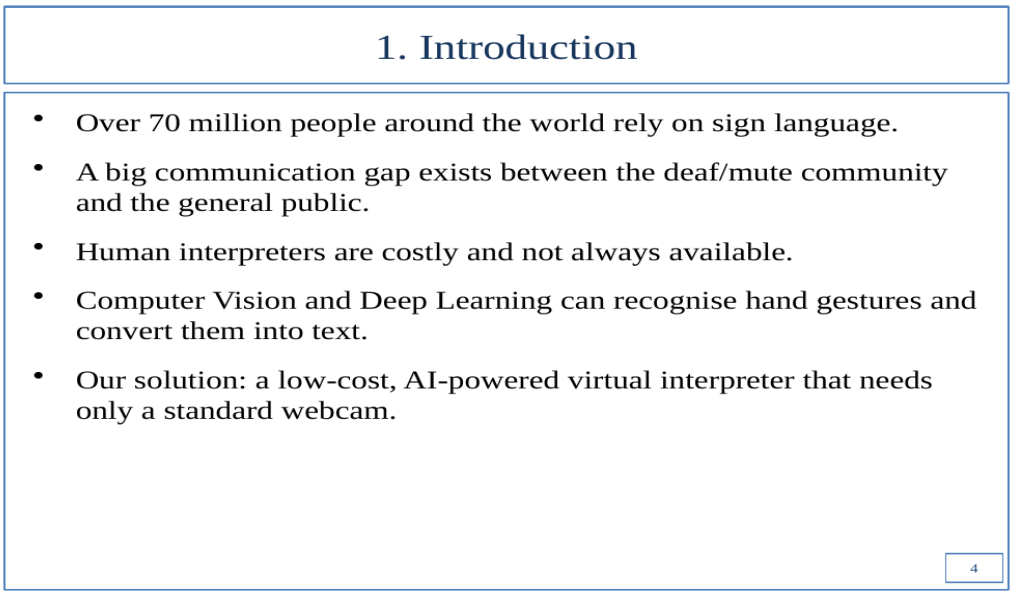
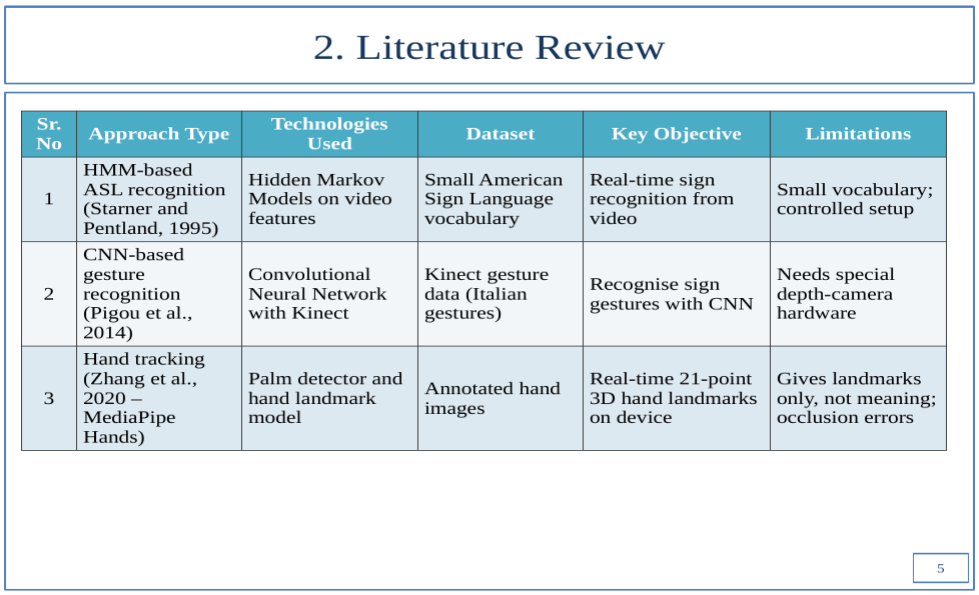


Fig. No. 7.4: Introduction Page

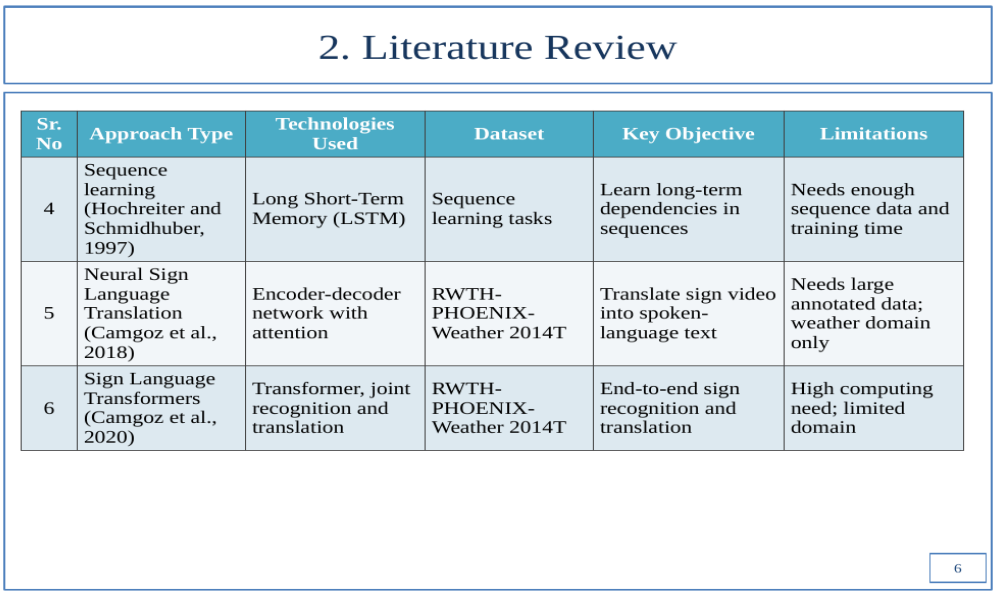


2. Literature Review

Sr. No	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	HMM-based ASL recognition (Starner and Pentland, 1995)	Hidden Markov Models on video features	Small American Sign Language vocabulary	Real-time sign recognition from video	Small vocabulary; controlled setup
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3	Hand tracking (Zhang et al., 2020 – MediaPipe Hands)	Palm detector and hand landmark model	Annotated hand images	Real-time 21-point 3D hand landmarks on device	Gives landmarks only, not meaning; occlusion errors

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Fig. No. 7.5: Literature Review



2. Literature Review

Sr. No	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
4	Sequence learning (Hochreiter and Schmidhuber, 1997)	Long Short-Term Memory (LSTM)	Sequence learning tasks	Learn long-term dependencies in sequences	Needs enough sequence data and training time
5	Neural Sign Language Translation (Camgoz et al., 2018)	Encoder-decoder network with attention	RWTH-PHOENIX-Weather 2014T	Translate sign video into spoken-language text	Needs large annotated data; weather domain only
6	Sign Language Transformers (Camgoz et al., 2020)	Transformer, joint recognition and translation	RWTH-PHOENIX-Weather 2014T	End-to-end sign recognition and translation	High computing need; limited domain

6

Fig. No. 7.6: Literature Review

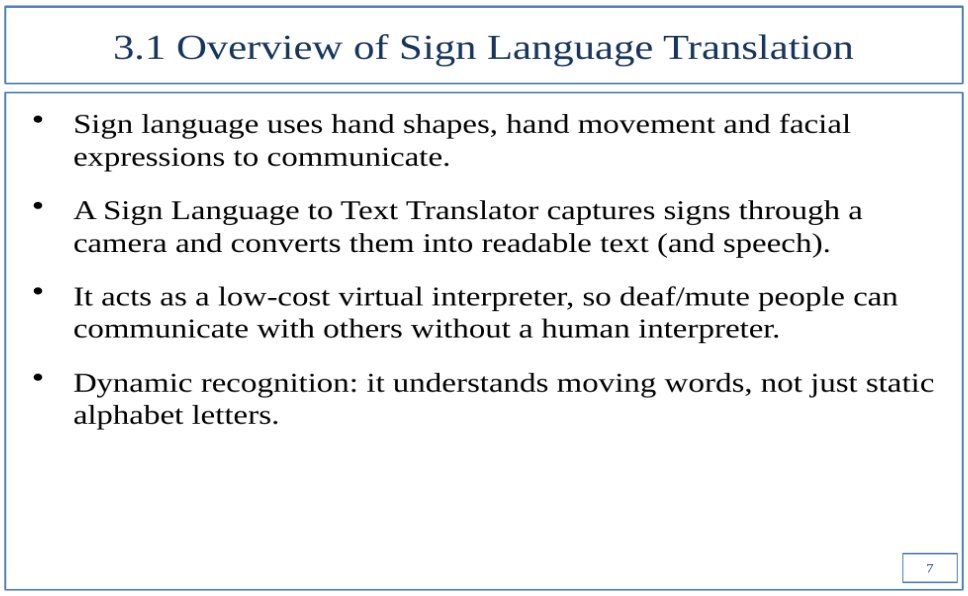


Fig. No. 7.7: Information To Seminar Topic – Overview

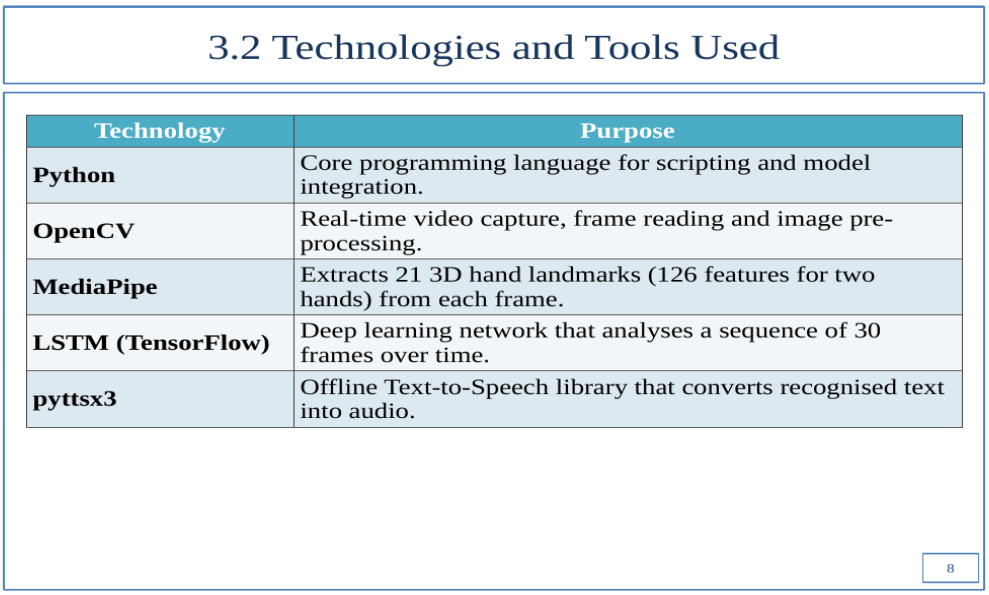


Fig. No. 7.8: Information To Seminar Topic – Technologies

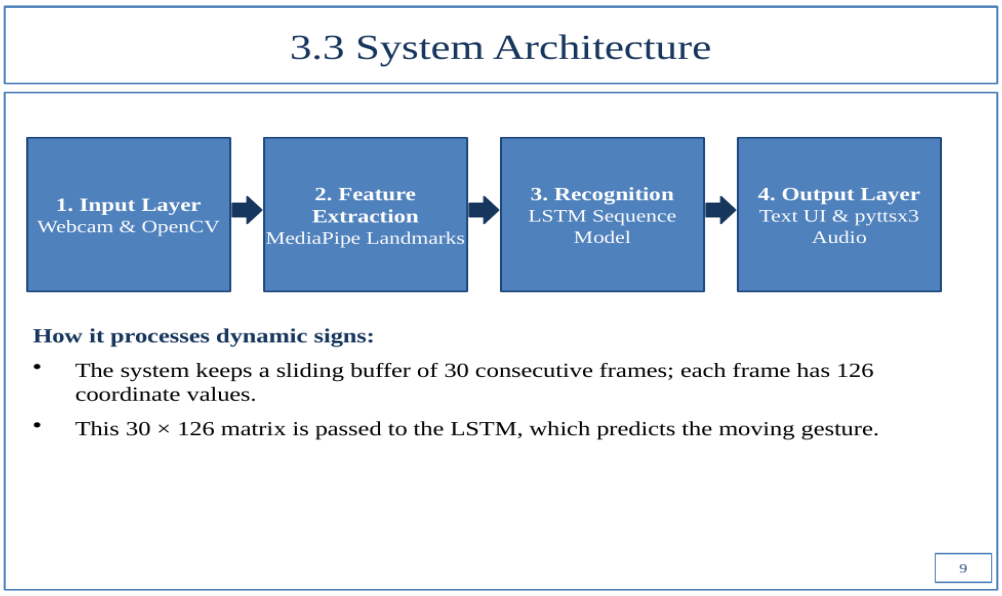


Fig. No. 7.9: Information To Seminar Topic – System Architecture

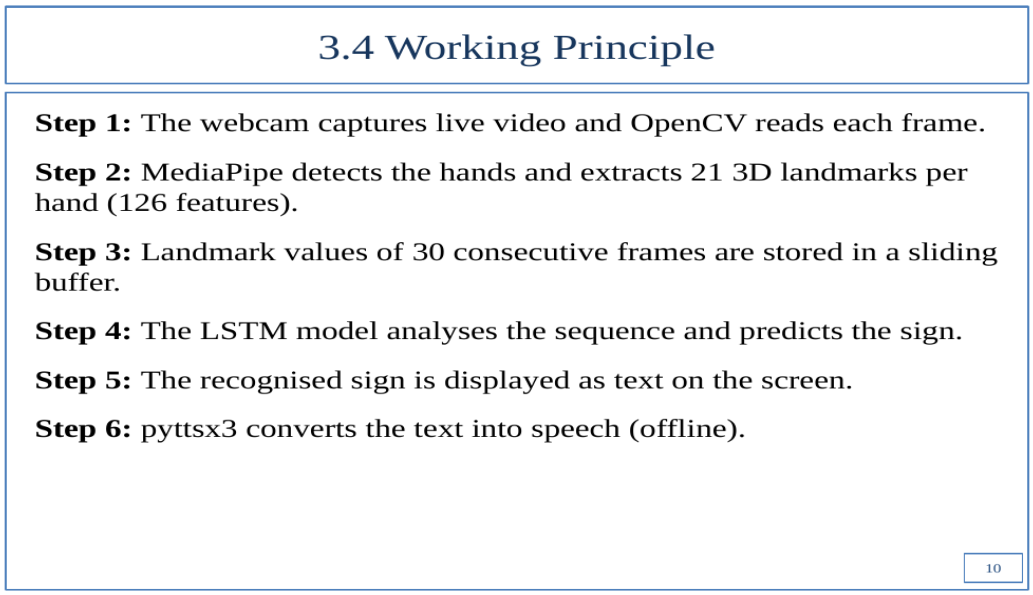


Fig. No. 7.10: Information To Seminar Topic – Working Principle

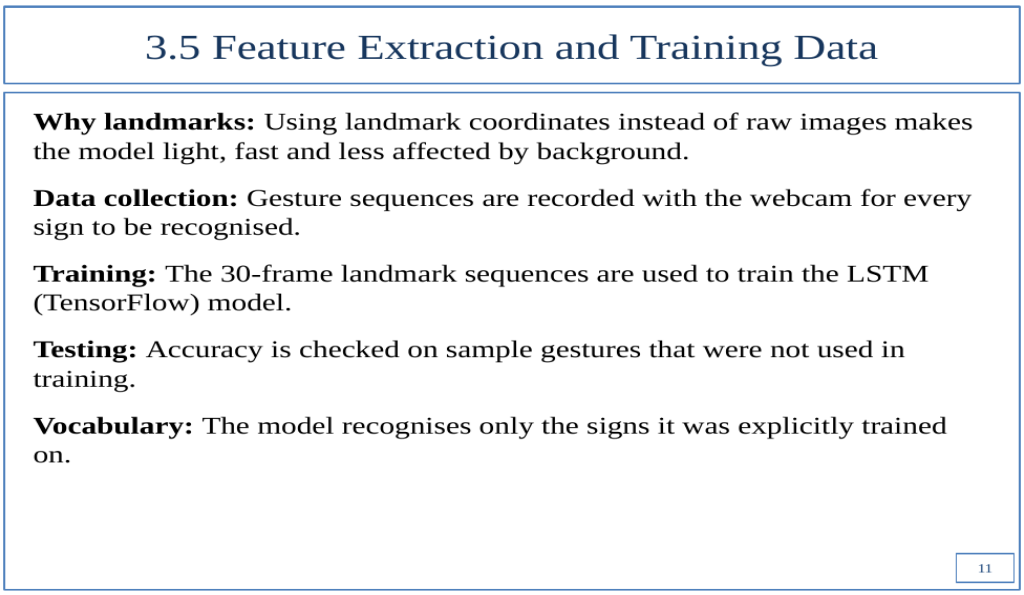


Fig. No. 7.11: Information To Seminar Topic – Feature Extraction

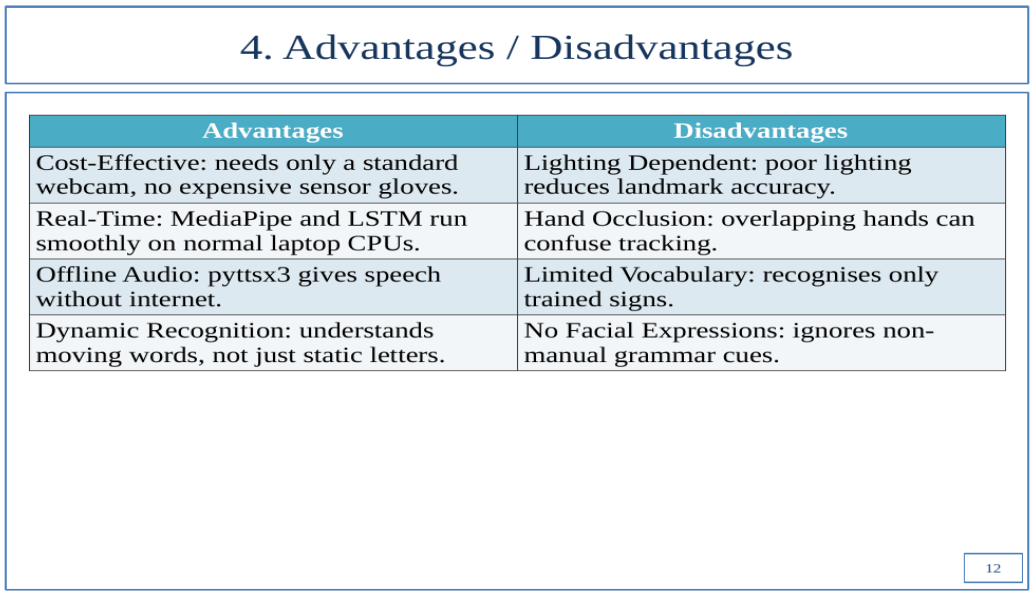


Fig. No. 7.12: Advantage And Disadvantage

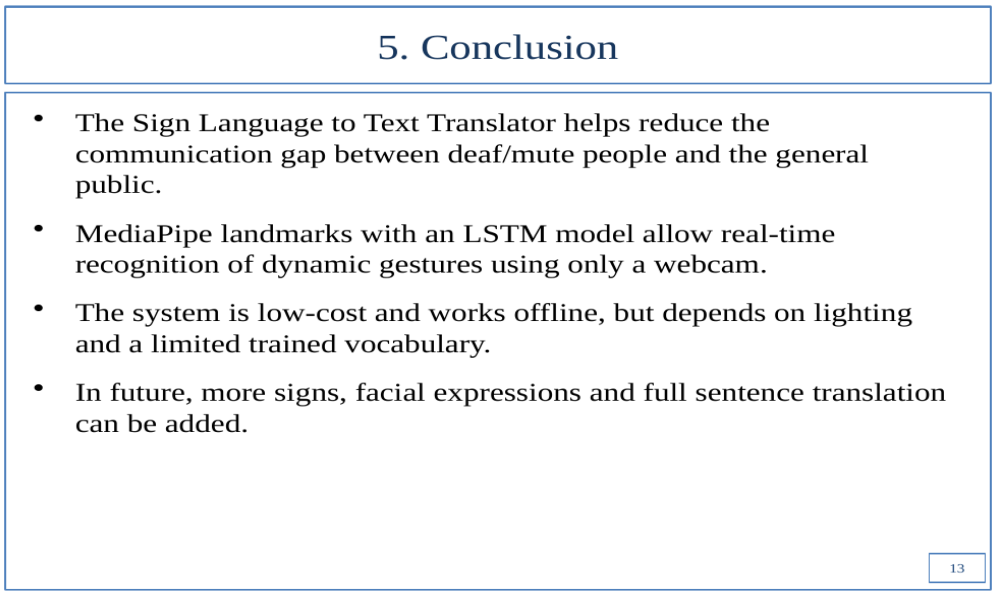


Fig. No. 7.13: Conclusion

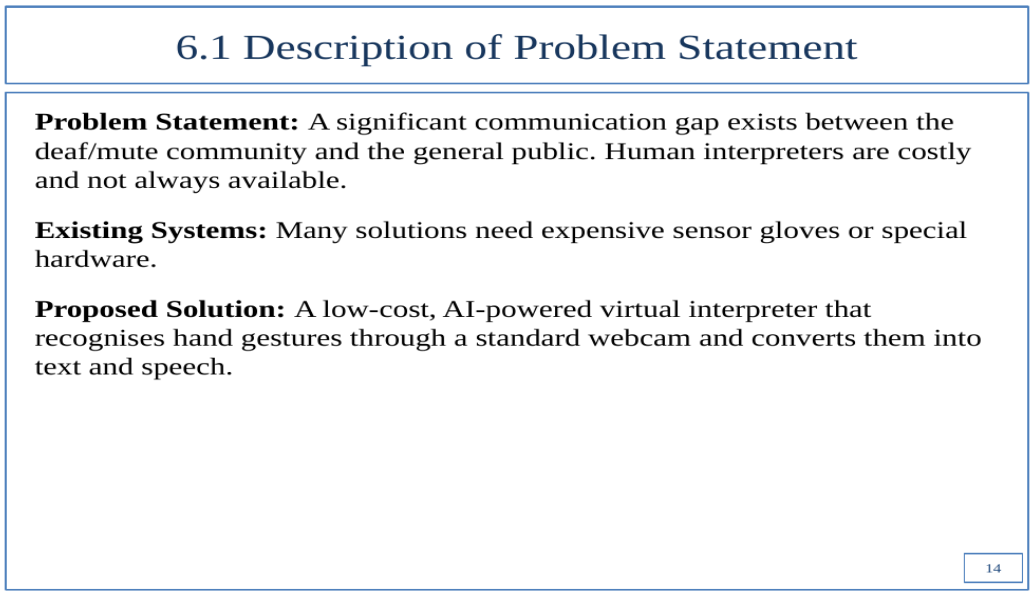


Fig. No. 7.14: Description of problem statement

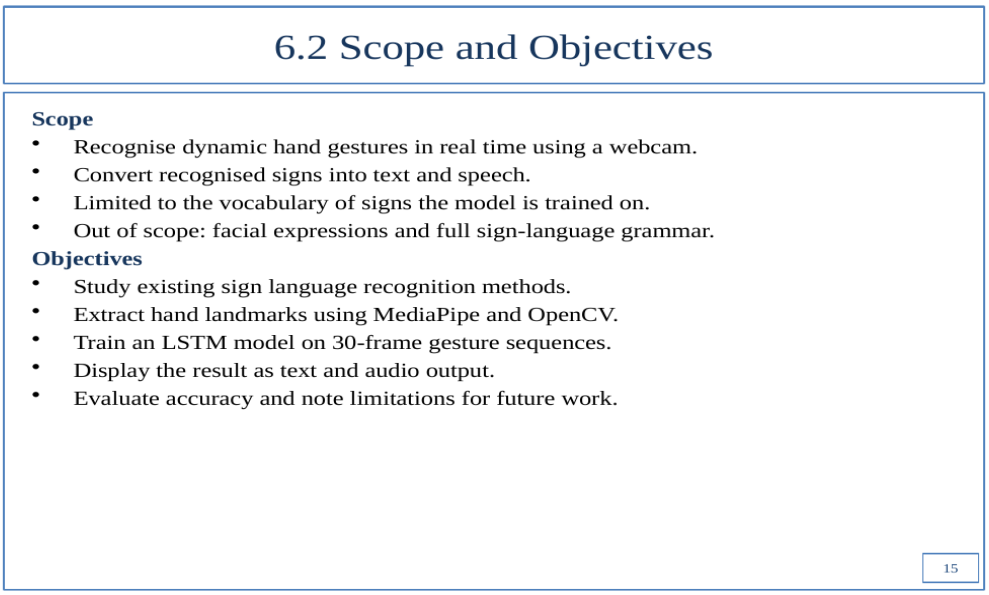


Fig. No. 7.15: Scope And Objectives

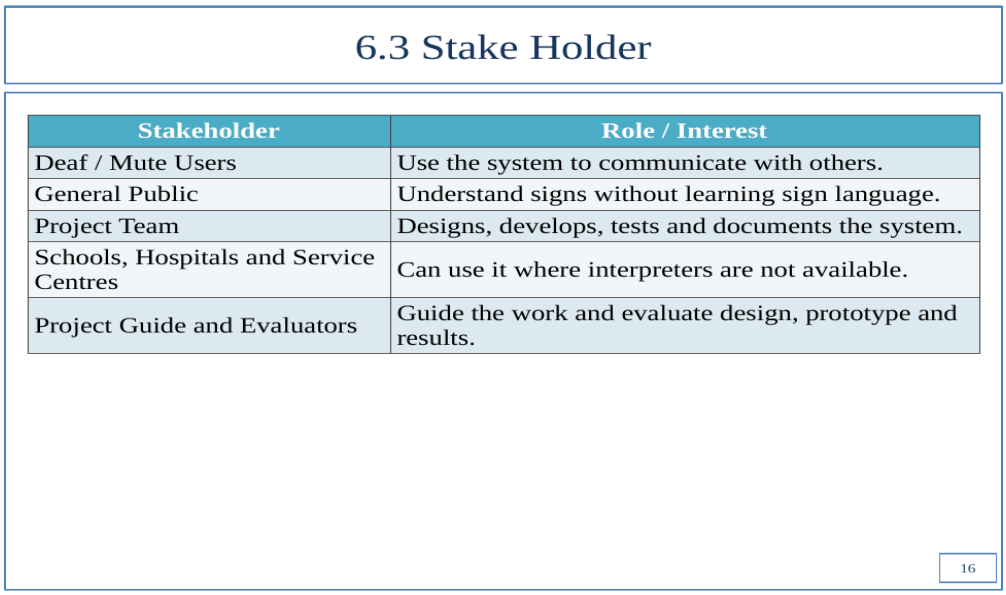


Fig. No. 7.16: Stake Holder

6.4 Seminar Execution Plan

Week	Activity	Expected Output
Week 1	Selection of seminar topic and finalising the title	Topic approval
Week 2	Literature survey of sign language recognition work	Literature review notes
Week 3	Study of tools: OpenCV, MediaPipe, TensorFlow, pyttsx3	Tool list
Week 4	Designing architecture and working flow	Architecture diagram
Week 5	Collecting gesture data and building the prototype	Working prototype
Week 6	Training the LSTM and testing accuracy	Test results
Week 7	Preparing the seminar report and slides	Report and PPT draft
Week 8	Final presentation and submission	Final submission

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Fig. No. 7.17: Seminar Execution Plan

6.5 Platform / Equipment / Resources

Programming: Python.

Computer Vision: OpenCV for video capture and pre-processing.

Hand Tracking: Google MediaPipe (21 hand landmarks).

Deep Learning: TensorFlow / Keras for the LSTM model.

Text-to-Speech: pyttsx3 (offline).

Data Handling: NumPy and Pandas.

Environment: VS Code, Jupyter Notebook or Google Colab.

Hardware: Laptop with webcam and at least 8 GB RAM.

Dataset: Gesture sequences recorded by the team for the chosen signs.

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Fig. No. 7.18: Platform/ Equipment/ Resources

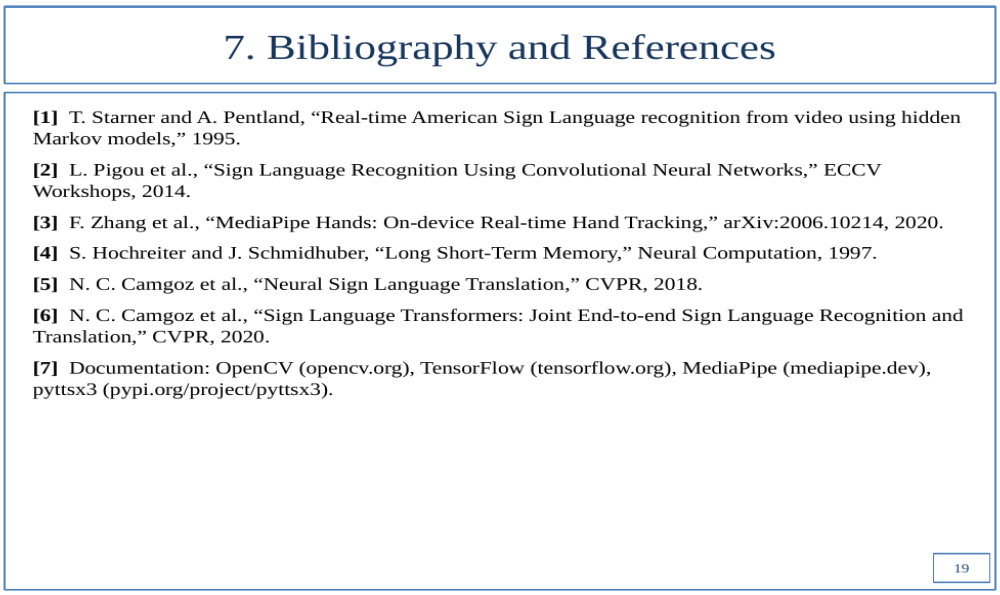


Fig. No. 7.19: Bibliography and References





BIBLIOGRAPHY

❖ **Books**

1. Deep Learning — Ian Goodfellow, Yoshua Bengio and Aaron Courville.
2. Learning OpenCV 3: Computer Vision in C++ with the OpenCV Library — Adrian Kaehler and Gary Bradski.

❖ **Websites**

1. https://opencv.org
2. https://www.tensorflow.org
3. https://mediapipe.dev
4. https://pypi.org/project/pyttsx3

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1. T. Starner and A. Pentland, “Real-time American Sign Language recognition from video using hidden Markov models,” 1995.

2. L. Pigou et al., “Sign Language Recognition Using Convolutional Neural Networks,” in ECCV Workshops, 2014.

3. F. Zhang et al., “MediaPipe Hands: On-device Real-time Hand Tracking,” arXiv preprint arXiv:2006.10214, 2020. https://arxiv.org/abs/2006.10214

4. S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

5. N. C. Camgoz, S. Hadfield, O. Koller, H. Ney and R. Bowden, “Neural Sign Language Translation,” in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2018.

6. N. C. Camgoz, O. Koller, S. Hadfield and R. Bowden, “Sign Language Transformers: Joint End-to-end Sign Language Recognition and Translation,” in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2020.

7. Documentation: OpenCV (opencv.org), TensorFlow (tensorflow.org), MediaPipe (mediapipe.dev), pyttsx3 (pypi.org/project/pyttsx3).

SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Name of student	Seat No
1	25610960009	Giridhar Fulsing Pawar	388607
2	25610960013	Dipak Motiram Gangurde	388611
3	25610960020	Harshad Shridhar Borase	388615
4	25610960001	Lalit Anil Patil	388600

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388607	388611	388615	388600
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	50				
	Scaled to 25					

Sign:

Name : Prof. N. D. Patel

(Course Expert/s)